

Logistic regression assumptions and diagnostics

Agenda and reminders

- + HW 2 due Friday
- + Department seminar tomorrow (9/11) at 11am in ZSR auditorium
 - + please refrain from wearing colognes, perfumes, and/or heavily scented body and hair products
- + Today: more on diagnostics

Shape assumption and quantile residuals

(In R)

Multicollinearity

What is multicollinearity?

Example: College scorecard data

The Department of Education compiles the College Scorecard, which is used to help prospective college students compare schools. For each school, variables include:

- + CONTROL: whether the school is public or private
- + SATVRMID: midpoint of SAT critical reading scores of students attending the school
- + ACTCMMID: midpoint of the ACT cumulative scores
- + UGDS: number of undergraduate students at the school
- + NPT4: average cost to attend the school
- + PCTFLOAN: fraction of undergraduates receiving a federal student loan
- + MD_EARN_WNE_P10: median salary of students 10 years after graduation

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Which of these variables may suffer from multicollinearity?

The impact of multicollinearity

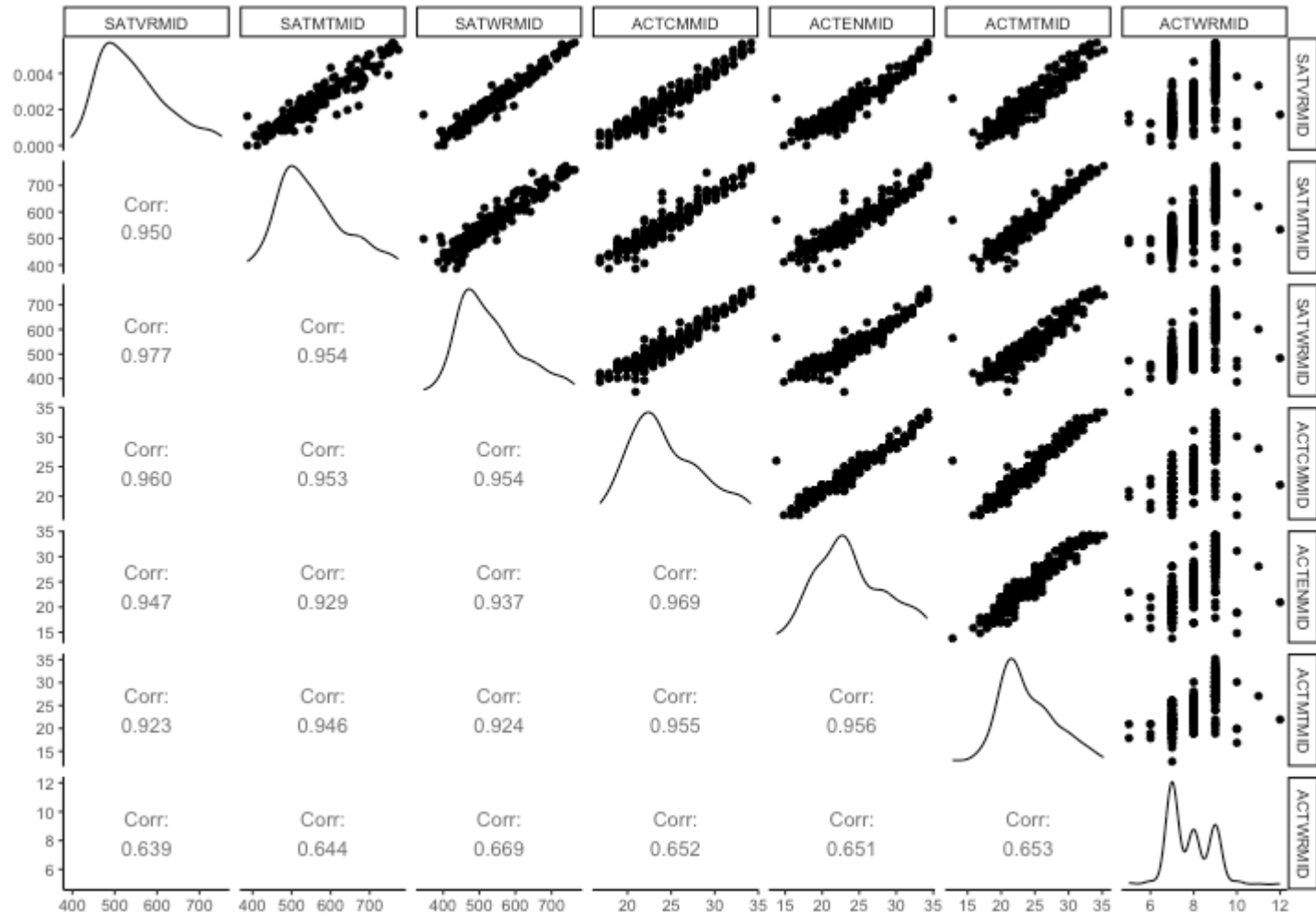
How does correlation between the explanatory variables impact the fitted model?

- + Increased variability in coefficient estimates
- + Difficulty interpreting coefficients

Diagnosing multicollinearity

How do I detect multicollinearity in my data?

Scatterplot and correlation matrix



Variance inflation factors

Variance inflation factors in R

```
library(car)
```

```
m1 <- glm(RPY_3YR_70 ~ CONTROL + SATVRMID +  
          SATMTMID + SATWRMID + ACTCMMID + ACTENMID +  
          ACTMTMID + ACTWRMID + UGDS +  
          PCTFLOAN + MD_EARN_WNE_P10 + NPT4,  
          data = scorecard, family = binomial)  
vif(m1)
```

##	CONTROL	SATVRMID	SATMTMID	SATWRMID	ACTCMMID
##	3.511039	14.752767	11.246146	13.763868	11.603111
##	ACTENMID	ACTMTMID	ACTWRMID	UGDS	PCTFLOAN
##	12.258720	8.837329	1.671265	2.426494	2.603111
##	MD_EARN_WNE_P10	NPT4			
##	1.333261	2.536284			

Addressing model issues

How should we handle multicollinearity in the explanatory variables? Discuss with a neighbor for a few minutes, then we will discuss as a group.

Outliers and influential points

What is an outlier?

Cook's distance

Addressing model issues

How should we handle outliers and influential points? Discuss with a neighbor for a few minutes, then we will discuss as a group.

Class activity

https://sta214-f25.github.io/class_activities/ca_08.html

At the end of class, render to HTML and submit HTML on Canvas